AQA 3.1.8 Thermodynamics: lattice enthalpies

AQA A-level Chemistry · Revision questions · Sheet 1 of 3

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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22 marks · about 25 minutes

Key information

- Lattice formation: $M^+(g) + X^-(g) \rightarrow MX(s)$, exothermic; lattice dissociation is the reverse and endothermic
- First IE: $X(g) \rightarrow X^+(g) + e^-$; first EA: $X(g) + e^- \rightarrow X^-(g)$ (exothermic)
- Larger lattice enthalpy: higher ionic charges and smaller ions give stronger attraction
- Perfect ionic model: ions are perfect spheres / point charges with no covalent character
- Born–Haber value bigger than perfect ionic value → covalent character (small, highly charged cation polarises the anion)

Section A: Short-answer questions

01 The enthalpy of lattice formation for caesium iodide obtained by experiment is −585 kJ mol^{-1}. The value obtained by calculation using the perfect ionic model is −582 kJ mol^{-1}. Deduce what these values indicate about the bonding in caesium iodide.

[1 mark]

02 Give an equation, including state symbols, for the process that represents the second electron affinity of oxygen.

[1 mark]

03 The Born–Haber and perfect ionic model values for the enthalpy of lattice formation of sodium chloride are very similar. State what this shows about the bonding in sodium chloride.

[1 mark]

04 Define the term *first ionisation energy*.

[2 marks]

05 The enthalpies of lattice formation of $MgCl_2$, MgO and $BaCl_2$ are −2018, −2493 and −3889 kJ mol^{-1} (not in order).

Match each substance to its enthalpy of lattice formation.

[1 mark]

06 Give an equation, including state symbols, for the process that represents the enthalpy of lattice dissociation of sodium oxide.

[1 mark]

07 Explain why the first electron affinity of chlorine is exothermic.

[1 mark]

08 Define the term *first electron affinity*.

[2 marks]

09 The table shows the theoretical lattice enthalpy, based on a perfect ionic model, and an experimental value for the enthalpy of lattice formation of silver chloride.

	Theoretical	Experimental
Enthalpy of lattice formation / kJ mol^{-1}	−770	−905

State why there is a difference between the theoretical and experimental values.

[1 mark]

Section B: Longer questions

10 This question is about silver iodide.

10.1 Define the term *enthalpy of lattice formation*.

[2 marks]

10.2 The enthalpy of lattice formation of silver iodide obtained from experimental data is –869 kJ mol^{-1}. A calculation based on a perfect ionic model gives a smaller numerical value. Explain this difference.

[2 marks]

11 This question is about the enthalpies of lattice formation of some Group 1 and Group 2 compounds.

11.1 Define the term *enthalpy of lattice formation*.

[2 marks]

11.2 Give an equation, including state symbols, for the enthalpy of lattice formation of potassium bromide.

[1 mark]

11.3 Predict whether potassium chloride has a more or less exothermic enthalpy of lattice formation than potassium bromide. Explain your answer.

[2 marks]

11.4 Explain why the enthalpy of lattice formation of calcium oxide is much more exothermic than that of potassium bromide.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	(Almost / mostly) purely / perfectly ionic	*Source: 7405/1 Jun 2019 Q01.3* If ionic not mentioned allow no / little covalent character; penalise atoms / molecules	1
02	$O^-(g) + e^- \rightarrow O^{2-}(g)$	*Source: Practice question (written for this set)*	1
03	It is (almost) purely ionic / little covalent character	*Source: Practice question (written for this set)*	1
04	M1 enthalpy change when one mole of electrons is removed from one mole of gaseous atoms M2 to form one mole of gaseous 1+ ions	*Source: Practice question (written for this set)* Allow equation $X(g) \rightarrow X^+(g) + e^-$ for M2	2
05	MgO −3889; $MgCl_2$ −2493; $BaCl_2$ −2018	*Source: 7405/1 Nov 2020 Q01.2* All three needed Original asks students to draw lines between columns	1
06	$Na_2O(s) \rightarrow 2Na^+(g) + O^{2-}(g)$	*Source: Practice question (written for this set)*	1
07	Energy is released as the added electron is attracted to the nucleus	*Source: Practice question (written for this set)*	1
08	M1 enthalpy change when one mole of gaseous atoms M2 each gain one electron to form one mole of gaseous 1– ions	*Source: Practice question (written for this set)*	2
09	(Has) covalent character / partial covalent bonding (as well as ionic)	*Source: 7405/1 Nov 2020 Q01.3* Allow chloride ion polarised / distorted Ignore not perfectly ionic; do not allow molecules	1
10.1	M1 enthalpy change when 1 mol of solid ionic compound / ionic lattice M2 is formed from its gaseous ions	*Source: 7405/1 Jun 2017 Q01.1, 01.3* Allow $M^+(g) + X^-(g) \rightarrow MX(s)$ CE = 0/2 if wrong process (e.g. lattice dissociation or formation)	2
10.2	M1 AgI contains covalent character / not completely ionic / ions distorted M2 forces / bonds holding the lattice together are stronger	CE = 0/2 if atoms / molecules Ignore electronegativity; stem gives the experimental value from the earlier part	2
11.1	M1 enthalpy change when 1 mol of solid ionic compound is formed M2 from its gaseous ions	*Source: Practice question (written for this set)*	2
11.2	$K^+(g) + Br^-(g) \rightarrow KBr(s)$		1
11.3	M1 more exothermic M2 Cl^- is smaller than Br^- so stronger attraction between ions		2
11.4	M1 Ca^{2+} and O^{2-} have higher charges (and are smaller) M2 stronger attraction between oppositely charged ions		2